## Appendix: Comparative Tool Matrix I

**Symbol key** (applies to all cells): **Y** = full native support · ∼ = partial or plugin-based · **N** = absent. See the comparative feature matrix in the Results section for a colour-coded heatmap rendering of this table.

# Appendix: Comparative Tool Matrix I

Table 1: Comparative feature matrix (14 capabilities × 10 tools). Y = full native support, ~ = partial or plugin-based, N = absent.

| Capability | `template` / | Snakemake | Nextflow | CWL | Quarto | Jupyter Book | R Markdown | DVC | Overleaf | OpenAI Prism |
|---|---|---|---|---|---|---|---|---|---|---|
| Pipeline orchestration | Y | Y | Y | Y | ~ | N | N | Y | N | N |
| Manuscript rendering | Y | N | N | N | Y | Y | Y | N | Y | Y |
| Testing enforcement | Y | N | N | N | N | N | N | N | N | N |
| Coverage thresholds | Y | N | N | N | N | N | N | N | N | N |
| Cryptographic provenance | Y | N | ~[1] | N | N | N | N | ~[2] | N | N |
| Steganographic watermarking | Y | N | N | N | N | N | N | N | N | N |
| Multi-project management | Y | N | N | N | N | N | N | N | ~ | ~ |
| AI-agent documentation | Y | N | N | N | N | N | N | N | ~ | ~ |

# Appendix: Comparative Tool Matrix I

| Capability | `template` / | Snakemake | Nextflow | CWL | Quarto | Jupyter Book | R Markdown | DVC | Overleaf | OpenAI Prism |
|---|---|---|---|---|---|---|---|---|---|---|
| Agentic skill protocol (SKILL.md / MCP) | Y | N | N | N | N | N | N | N | N | N |
| Interactive TUI | Y | N | N | N | N | N | N | N | N | N |
| Zero-mock policy | Y | N | N | N | N | N | N | N | N | N |
| Container support | ~ | Y | Y | Y | N | N | N | N | N | N |
| Distributed execution | N | Y | Y | Y | N | N | N | ~[3] | N | N |
| Multi-language (R/Julia) | N | Y | N | Y | Y | Y | Y | Y | N | N |

## Appendix: Comparative Tool Matrix I

[1] Nextflow lineage records operate at workflow scope rather than as a rendered-document citation graph. [2] DVC provides content-addressed versioning for data artifacts via its object store. [3] DVC integrates with remote storage (S3, GCS, Azure) but does not natively orchestrate distributed compute. This matrix is a versioned manuscript snapshot, not a continuously updated product survey. A future refresh should re-check every external capability against current primary documentation before changing a cell.